

VINAY DILIP JAYASELAN

Energy Engineer | Digital Twin & IoT | BIM/VDC | R&D Project Management

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PROFESSIONAL SUMMARY

Recent MSc graduate in Energy-Efficient Building Design (Lund University, 2026) with hands-on R&D experience developing IoT-enabled digital twin platforms for real-time building performance monitoring and energy optimisation. Brings five years of BIM/VDC engineering experience delivering complex, cross-functional projects including hospitals, data centres, pharmaceutical facilities, and battery-manufacturing plants in international hybrid teams. Known for a structured, data-driven approach to coordination and technical delivery, strong communication skills, and the ability to manage multidisciplinary stakeholders across geographies.

EDUCATION

Ms. Energy-Efficiency in Building Design

Graduating: 2026

Lund University, Lund, Sweden

Thesis: Developing a Cost-Effective Digital Twin Framework for Energy-Efficient Operation in Existing Buildings

- Designed and independently delivered an end-to-end IoT-enabled digital twin platform, including requirement scoping, sensor network deployment, MQTT integration, SQL database setup, Python-based analytics, and real-time dashboard visualisation for building performance monitoring.
- Managed the full project lifecycle from concept to implementation, ensuring technical outputs aligned with research objectives and stakeholder expectations, directly reflecting R&D project management capability.
- Demonstrated the ability to work across building physics, data engineering, and software integration, synthesising these components into a coherent, decision-ready system.

Relevant coursework: Building Energy Design • HVAC • Solar Energy • LCA/LCC • Moisture & Heat Analysis • Lighting • Material Physics

Bachelor of Architecture

Graduated: 2019

VTU, India | Architectural Design • Urban Design • Structures & Installations • Construction

Diploma in Civil Engineering

Graduated: 2013

RJS Polytechnic, India | Construction • Structures & Installations • Estimation & Quantity

PROFESSIONAL EXPERIENCE

VDC – BIM Engineer

January 2022 – April 2025

Doxel.ai → ModelyzeLabs.ai (internal restructuring), California, USA (Hybrid)

- Led end-to-end VDC delivery for hospitals, pharmaceutical facilities, data centres and battery-manufacturing plants, coordinating multidisciplinary and geographically distributed teams under tight schedules and technical constraints.
- Drove adoption of BIM Execution Plans (BEPs) and technology-enabled workflows, measurably reducing rework and improving model accuracy and constructability across project types.
- Managed 3D model development, version control, and clash detection using Revit, Navisworks, Colibri, and Dynamo, ensuring alignment across architectural, structural, and MEP disciplines.
- Planned, coordinated, and tracked project tasks, resources, and milestones; provided regular progress updates to project managers and client stakeholders across time zones.
- Identified technical risks in project execution and contributed to mitigation strategies, maintaining delivery targets against defined scope and timelines.

- Applied construction methodology knowledge to optimise VDC processes, including BIM-based quantity takeoffs, cost estimation, and schedule integration

BIM Architect

October 2020 – December 2021

Girish M Swamy Consultants Pvt. Ltd, Bangalore, India

- Orchestrated multidisciplinary coordination across architectural, structural, and MEP disciplines for major hospital and educational institution projects.
- Produced and managed 2D construction drawings from BIM models; delivered client presentations at project milestones and coordinated on-site with PHE, Electrical, and Fire teams.

Architect

August 2019 – September 2020

Rathi Associate, Bangalore, India

- Produced high-quality Revit and AutoCAD deliverables, including detailed designs, schedules, and building-services documentation.
- Coordinated cross-disciplinary workflows and resolved technical issues directly with contractors, consultants, and clients.
- Delivered design documentation for diverse projects, including commercial buildings, residential apartments, and township planning.

Architect - Intern

July 2018 – July 2019

B+A Architects, Bangalore, India

- Supported design and construction phases for national-level projects, preparing precise drawing layouts and coordinating directly with clients.
- Delivered technical detailing, research, and documentation across diverse sectors, including F&B, residential, retail, and commercial projects.

Architect - Intern

June 2014 – March 2018

Citati Architects, Bangalore, India

(Internship completed alongside Bachelor of Architecture studies)

- Produced construction drawings, architectural details, and 3D models aligned with project requirements.
- Assisted small-scale projects - including residential, schools, and public buildings - with documentation and visual illustrations.

KEY SKILLS

Energy & Simulation: Building energy modelling, HVAC design, WUFI, EQUA IDA ICE, CFD, Ansys Fluent, LCA/LCC, solar energy analysis.

Digital Twin & IoT: Sensor integration, MQTT, SQL databases, Python-based analytics, real-time dashboard visualisation.

BIM / VDC: Revit, Navisworks, Rhino, Rhino.Inside, Grasshopper, Dynamo, AutoCAD.

Data & Programming: Python, SQL, JavaScript, Power BI, AWS, Linux.

Project Management: Cross-functional team coordination, milestone tracking, risk identification, stakeholder communication.

Languages & Intercultural: English (Professional), Swedish (Studying), Swedish B-Körkort / Driver's License (In progress) • International experience via MSc studies in Sweden and hybrid work with US-based teams.

CERTIFICATIONS

Revit Architecture • AutoCAD • Python • Rhino.Inside • Grasshopper • EQUA IDA ICE • Power BI